

Viscous polyconvex model: calibration and numerical example

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1 Calibration

Experimental data. The model is calibrated against the uniaxial tests on VHB 4910 reported in [1], digitised with [2]. Two datasets are used: (i) two quasi-static loading curves up to $\lambda = 5$ (22 points), which characterise the equilibrium response, and (ii) eleven single loading–unloading cycles at constant stretch rates $\dot{\lambda} \in \{0.01, 0.03, 0.05\} \text{ s}^{-1}$ and maximum stretches $\lambda_{\max} \in \{1.5, 2.0, 2.5, 3.0\}$ (1158 points), with test durations between 17 s and 357 s.

Model and test kinematics. The generalised Maxwell model combines a neo-Hookean equilibrium term, with shear modulus μ_e , and n viscous polyconvex branches, each characterised by a shear modulus μ_i and a relaxation time τ_i . Every test is reproduced at a single material point under homogeneous incompressible uniaxial tension, $\mathbf{F} = \text{diag}(\lambda, \lambda^{-1/2}, \lambda^{-1/2})$, with the hydrostatic pressure fixed by the traction-free lateral faces; the predicted nominal stress P_{11} is compared with the measured one. The internal variables are integrated with the time step $\Delta t = \Delta\lambda/\dot{\lambda}$ dictated by the stretch rate, using the *same* constitutive routines and return mapping (HyperFEM) as the finite element simulation of Section 2. No constitutive law is re-implemented between calibration and simulation.

Two-step identification. The parameters minimise the normalised mean squared error over the M curves of each dataset,

$$\mathcal{L}(\mathbf{p}) = \frac{1}{M} \sum_{k=1}^M \frac{1}{N_k} \sum_{j=1}^{N_k} \left(\frac{P_{k,j}^{\text{exp}} - P_{k,j}(\mathbf{p})}{\max_j |P_{k,j}^{\text{exp}}|} \right)^2, \quad (1)$$

in two sequential steps: **Step 1**, μ_e is fitted to the quasi-static curves (Nelder–Mead); **Step 2**, with μ_e fixed, the branch parameters $\{\mu_i, \tau_i\}_{i=1}^n$ are fitted simultaneously to the eleven cycles with a gradient-free, global particle swarm optimiser [3]. The seed of the random number generator is fixed in the script, so that both the swarm and the sampling below return the same values in every run.

Uncertainty. With the residual Jacobian $\mathbf{J} = \partial \mathbf{r} / \partial \mathbf{p}$ (central differences), the parameter covariance is estimated by the cluster-robust (sandwich) estimator [4]

$$\mathbf{\Sigma} = \frac{M}{M-1} (\mathbf{J}^T \mathbf{J})^{-1} \left(\sum_{k=1}^M \mathbf{J}_k^T \mathbf{r}_k \mathbf{r}_k^T \mathbf{J}_k \right) (\mathbf{J}^T \mathbf{J})^{-1}, \quad (2)$$

where residuals are clustered by experimental curve to account for their correlation along each curve. Table 1 reports the half-width of the 95% confidence interval (margin), the relative error (margin/estimate) and the normalised sensitivity $S_j = 2(\mathbf{J}^T \mathbf{J})_{jj} p_j^2 / \text{SSE}$, such that a relative perturbation ε of p_j increases the sum of squared errors by a factor $1 + \frac{1}{2} S_j \varepsilon^2$.

Table 1: Calibrated parameters of VHB 4910 (two-branch model). Coefficients of determination: $R^2 = 0.995$ (step 1) and $R^2 = 0.978$ (step 2).

Parameter	Estimate $\pm$ margin (95%)	Rel. error [%]	Sensitivity
<i>Step 1: Equilibrium term (neo-Hookean)</i>			
μ_e [kPa]	13.4 ± 0.09	0.7	1278.6
<i>Step 2: Non-equilibrium terms (neo-Hookean branches)</i>			
μ_1 [kPa]	11.1 ± 2.9	26.1	45.9
τ_1 [s]	152 ± 80	52.7	5.3
μ_2 [kPa]	30.5 ± 3.6	11.7	19.8
τ_2 [s]	7.07 ± 2.0	28.0	13.0

Results. The identification returns two branches with well separated relaxation times, a slow one ($\tau_1 = 152$ s) and a fast one ($\tau_2 = 7.1$ s), which span the time scales probed by the tests (Table 1). Figure 1 shows that the model captures the rate dependence, the hysteresis and the residual stretch after unloading, both across loading rates and across stretch amplitudes. The largest deviation, about 10% of the peak stress at $\lambda_{\max} = 2.5$, is attributable to the neo-Hookean equilibrium term, deliberately kept to limit the model to five parameters.

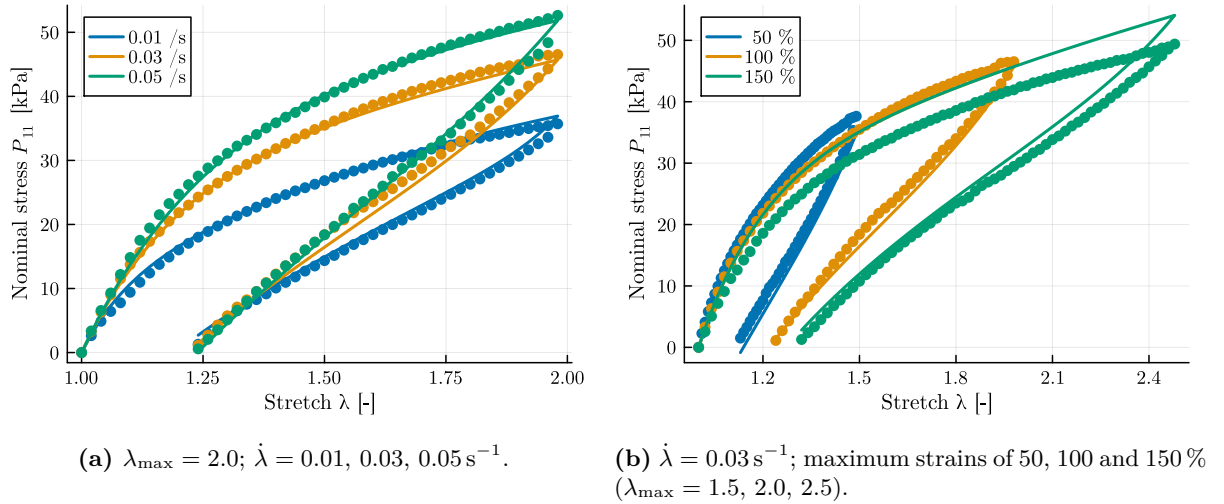


Figure 1: Calibrated model (lines) versus experimental data [1] (markers) for uniaxial loading–unloading cycles of VHB 4910.

Number of branches. Figure 2 propagates the parameter uncertainty to the predicted response: 100 parameter sets are sampled from $\mathcal{N}(\mathbf{p}^*, \mathbf{\Sigma})$ and evaluated on the test at $\lambda_{\max} = 2.0$ and $\dot{\lambda} = 0.03 \text{ s}^{-1}$. With two branches the ensemble stays within a narrow band around the measured cycle. Adding a third branch leaves the fit unchanged ($R^2 = 0.977$ in both cases, loss reduction of 0.1%) but scatters the ensemble over the whole stress range: the relative errors of the relaxation times grow from 28–53% to 438% and 1180%, and the swarm returns a negative shear modulus for one of the branches, outside the admissible range of the model. The additional parameters are therefore not identifiable from the available data, and the two-branch model is the richest one supported by the experiments.

Reproducibility. The whole section is produced by the script `calibration/Calibration.jl` of the repository [7] from the CSV data distributed with it, using Julia 1.12, HyperCalibra-

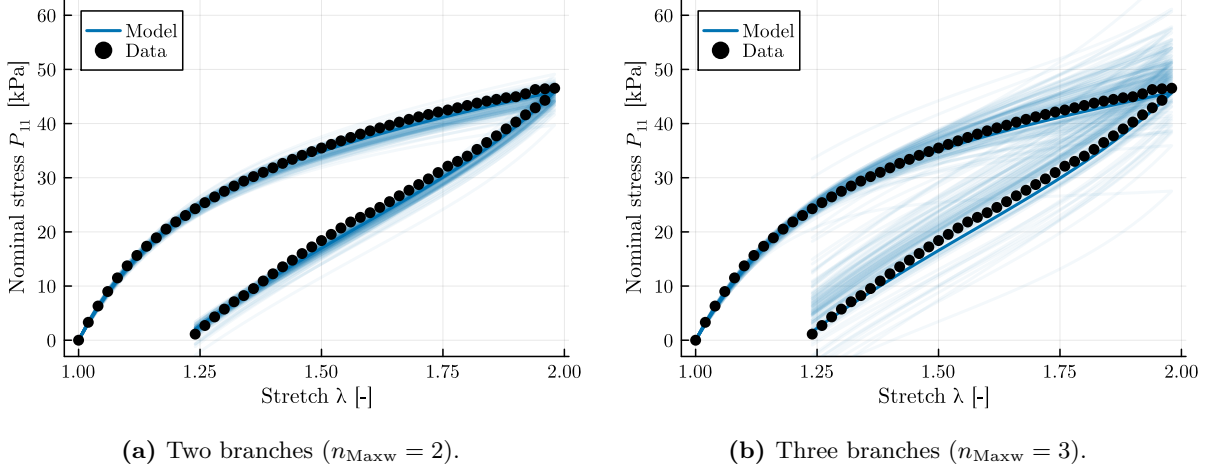


Figure 2: Propagation of the parameter uncertainty (100 samples, shaded) for $\lambda_{\text{max}} = 2.0$ and $\dot{\lambda} = 0.03 \text{ s}^{-1}$; optimal model (solid line) and experimental data (markers).

tion.jl v0.0.3 and HyperFEM.jl v0.0.6 [6]. It runs in about a minute on a laptop and returns the values of Table 1 in every run.

2 Numerical example

Problem set-up. A square membrane of side $w = 200 \text{ mm}$ and thickness $h = 1 \text{ mm}$, $\Omega_0 = (-w/2, w/2)^2 \times (0, h)$, with density $\rho_0 = 960 \text{ kg/m}^3$, is described by the two-branch model of Table 1, used as is. Near-incompressibility is enforced by the volumetric term $U(J) = \frac{\kappa}{2}(J - 1)^2$ with $\kappa = 2.5 \text{ MPa}$, i.e. $\kappa \approx 45 \mu_0$ with $\mu_0 = \mu_e + \mu_1 + \mu_2 = 55.0 \text{ kPa}$ the instantaneous shear modulus; $|J - 1| < 6 \cdot 10^{-4}$ throughout the simulation. No displacement constraint is prescribed (Figure 3): the membrane is free-floating, so that no work is exchanged through reactions and the energy balance involves the applied load only. The load is an out-of-plane traction $\bar{\mathbf{t}} = \bar{t}_0 f(t) \mathbf{e}_3$, $\bar{t}_0 = 20 \text{ Pa}$, applied on the six faces of the central element, a $w/7 \times w/7 \times h$ block (peak resultant force $\approx 35 \text{ mN}$). The pulse $f(t)$ is triangular: it grows linearly from 0 to 1 at $t = 0.05 \text{ s}$ and vanishes for $t \geq 0.1 \text{ s}$, after which the membrane evolves freely.

Discretisation. The domain is discretised with $7 \times 7 \times 1$ hexahedra with quadratic Lagrangian interpolation (2025 degrees of freedom) and Gauss quadrature exact for polynomials of degree 4. Time integration is implicit with $\Delta t = 1 \text{ ms}$ up to $t = 1 \text{ s}$ (1000 steps). With the velocity update $\frac{1}{2}(\mathbf{v}_n + \mathbf{v}_{n+1}) = (\mathbf{u}_{n+1} - \mathbf{u}_n)/\Delta t$, each step solves

$$\int_{\Omega_0} \rho_0 \frac{\mathbf{v}_{n+1} - \mathbf{v}_n}{\Delta t} \cdot \delta \mathbf{u} \, dV + \int_{\Omega_0} \mathbf{P}_{n+1/2} : \nabla_0 \delta \mathbf{u} \, dV = \int_{\Gamma_c} \bar{\mathbf{t}}(t_{n+1/2}) \cdot \delta \mathbf{u} \, dA, \quad (3)$$

with $\mathbf{P}_{n+1/2} = \frac{1}{2}(\mathbf{P}_n + \mathbf{P}_{n+1})$, where $\mathbf{P}_{n+1} = \partial_{\mathbf{F}} \Psi(\mathbf{F}_{n+1}; \mathbf{F}_n, \mathcal{A}_n)$ is the algorithmic first Piola–Kirchhoff stress and $\mathcal{A}_n$ the internal variables of the branches, updated by the return mapping at the end of each step. The nonlinear system is solved by Newton–Raphson with a direct solver (absolute and relative tolerances 10^{-8}) within Gridap [5].

Energy balance. The kinetic energy K , the stored energy $\mathcal{E}$, the accumulated viscous dissipation D and the external work W_{ext} ,

$$\begin{aligned} K &= \int_{\Omega_0} \frac{1}{2} \rho_0 |\mathbf{v}|^2 \, dV, & \mathcal{E} &= \int_{\Omega_0} \Psi(\mathbf{F}, \mathcal{A}) \, dV, \\ D &= \int_0^t \int_{\Omega_0} \mathcal{D} \, dV \, ds, & W_{\text{ext}} &= \sum_n \int_{\Gamma_c} \bar{\mathbf{t}}(t_{n+1/2}) \cdot (\mathbf{u}_{n+1} - \mathbf{u}_n) \, dA, \end{aligned} \quad (4)$$

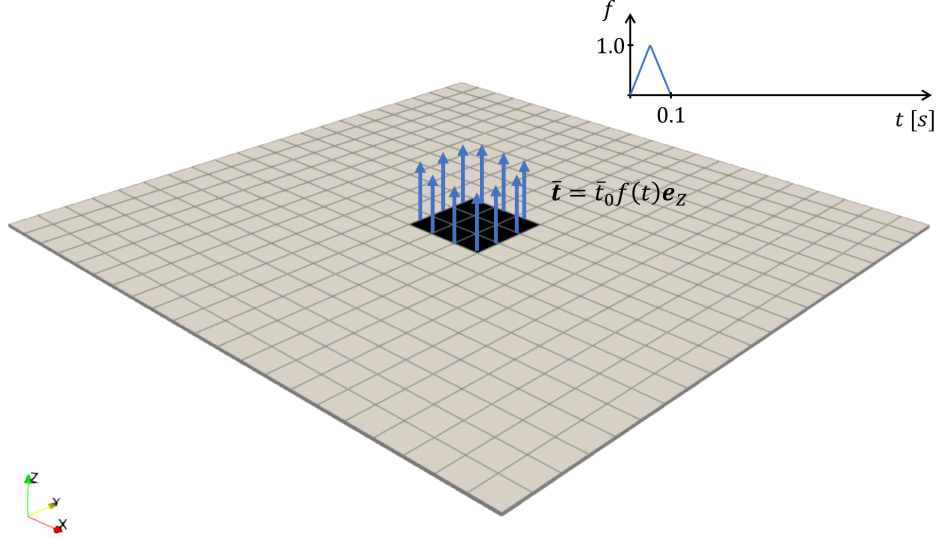


Figure 3: Free-floating square membrane: geometry, $7 \times 7 \times 1$ mesh, loaded central patch and time evolution $f(t)$ of the applied traction.

are monitored at every step and must satisfy $K + \mathcal{E} + D = W_{\text{ext}}$. Figure 4 shows that the pulse contributes $W_{\text{ext}} = 220 \mu\text{J}$ during $t \leq 0.1$ s. Afterwards the membrane is isolated: energy is continuously exchanged between its kinetic and elastic parts, while the dissipation grows monotonically, up to $D = 9.1 \mu\text{J}$ at $t = 1$ s, 4% of the injected energy. The balance residual $|K + \mathcal{E} + D - W_{\text{ext}}|$ grows linearly in time and stays below $2.9 \mu\text{J}$, i.e. below 1.3% of the injected energy, over the 1000 steps. It is a discretisation error: halving the time step reduces it to $0.95 \mu\text{J}$, while the dissipation itself changes by 1%.

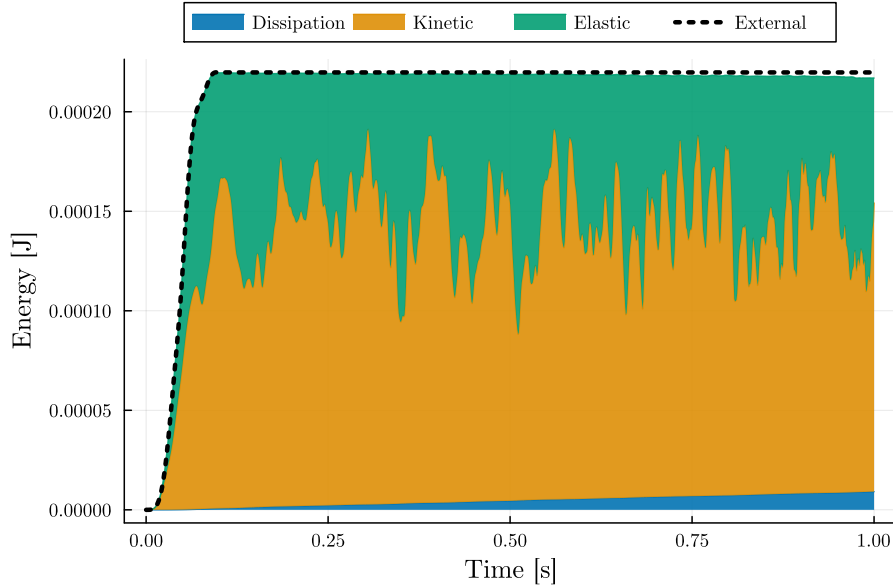


Figure 4: Energy balance: accumulated dissipation D , kinetic energy K and stored energy $\mathcal{E}$ (stacked areas) versus the external work W_{ext} (dotted line). The discrepancy of the total energy is a discretisation error.

Deformation. Figure 5 shows the deformed configuration at selected instants. The pulse bends the membrane around the loaded patch; the flexural wave reaches the free edges right

after the pulse ends ($t \approx 0.1$ s), the edges oscillate with a deflection of the order of 7 mm about the mean plane. The relaxation times of the material are much longer than the simulated time, so this oscillation is barely damped. The whole motion is superposed on the rigid-body translation induced by the impulse of the load, with a mean velocity of 4.5 cm/s.

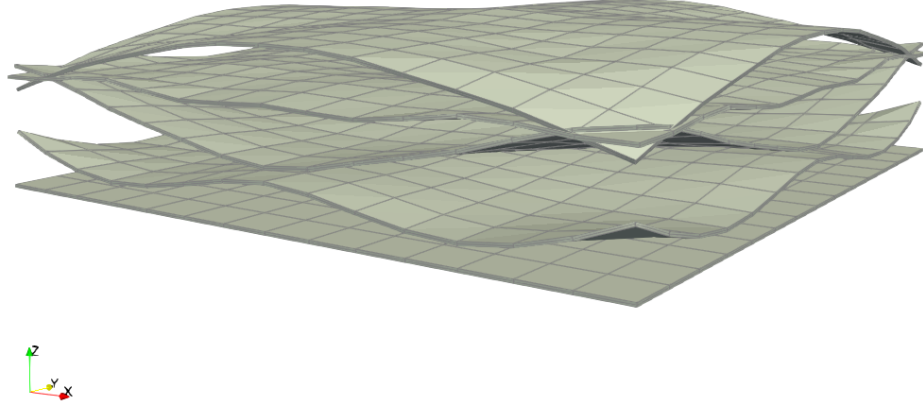


Figure 5: Deformed configuration at $t = 0, 0.15$ and 0.3 s.

Reproducibility. The example is produced by the script `simulation/ViscoTimeIntegrator-Neumann.jl` of the repository [7], with Julia 1.12, HyperFEM.jl v0.0.6 and Gridap v0.19. It writes the VTK snapshots and the energy history, and runs in about ten minutes on a laptop; two independent runs reproduce the energy history of Figure 4 to the last bit.

References

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- [6] HyperFEM.jl, <https://github.com/MultiSimOLab/HyperFEM.jl>; HyperCalibration.jl, <https://github.com/miguelmaso/HyperCalibration.jl>.
- [7] <https://github.com/miguelmaso/2026-ViscousPolyconvex>.